

Git Operations & Hands-On Labs

Bishtify – Build Skills, Not Just Resumes

■ Git Basics (Beginner Series)

1. Git Setup & First Repo

Operations

- git --version
- git config --global user.name
- git config --global user.email
- git init
- git status

Lab

- Create a local project folder
- Initialize Git
- Add a file and check status

2. Tracking Files & Commits

Operations

- git add
- git commit -m
- git log
- git diff

Lab

- Create 3 files
- Modify them
- Commit in stages
- Show history and diffs

3. Ignoring Files

Operations

- .gitignore

Lab

- Ignore .env, node_modules, *.log
- Prove ignored files don't get staged

■ Branching & Collaboration

4. Branching Basics

Operations

- git branch
- git checkout -b
- git switch

- git merge

Lab

- Create feature-login branch
- Make changes
- Merge into main

5. Merge Conflicts

Operations

- Conflict resolution
- git status
- Manual conflict fix

Lab

- Edit same line in two branches
- Trigger conflict
- Resolve and commit

6. Remote Repositories (GitHub)

Operations

- git remote add origin
- git push
- git pull
- git clone

Lab

- Push local repo to GitHub
- Clone it in a new folder
- Make changes and sync

■ Intermediate Git

7. Undoing Mistakes

Operations

- git restore
- git reset --soft
- git reset --hard
- git revert

Lab

- Undo staged file
- Remove last commit safely
- Revert a bad commit

8. Stashing Work

Operations

- git stash
- git stash pop

- git stash list

Lab

- Pause unfinished work
- Switch branch
- Resume work

9. Rewriting History

Operations

- git rebase
- git rebase -i

Lab

- Squash multiple commits
- Clean messy commit history

■ Real-World Git Workflows (High-Value Content)

10. Team Workflow Simulation

Lab

- Simulate 2 developers using branches
- Feature → PR → Merge
- Handle conflict like a real team

11. Release Workflow

Operations

- git tag
- git checkout tags

Lab

- Tag v1.0
- Create hotfix branch
- Release v1.1

12. Git for DevOps Engineers

Lab

- Store Terraform code in Git
- Use branches for environments (dev, prod)
- Roll back infra using Git history